

Dhruva Devaraaj

Vancouver, BC | +1 (672) 667-0768 | dhruvadevaraaj@gmail.com | [dhruva36.github.io](https://github.com/dhruva36) | github.com/dhruva36

EDUCATION

University of British Columbia

Aug. 2022 – May 2027

BA in Honours Economics, Concentrations in Math and Computer Science

Vancouver, BC

- **Awards:** Dean's Scholar, Work Learn International Undergraduate Research Award (\$8,700)
- **Honours Thesis:** Abortion Bans, Distance, and Maternal Morbidity After Dobbs
- **Relevant Coursework:** Advanced Econometrics, Methods of Empirical Research, Computational Macroeconomics, Game Theory, Calculus (I-III), Development Economics, Mathematical Proofs, Linear Algebra, Machine Learning & Data Mining, Probability Theory, Theory of Computing, Data Structures & Algorithms

EXPERIENCE

Research Assistant | *Vancouver School of Economics*

May 2026 – Present

- Assisting **Dr. Munir Squires** on a project studying the effects of heat exposure on mortality and longevity
- Developing a structural mortality model based on a first-hitting-time framework
- Applying the model to a 247,000-person longitudinal genealogical panel in Python and Julia, building the simulation pipeline used to confirm the estimator recovers known parameters before taking it to real data

Research Assistant | *Stanford University*

Nov. 2025 – Apr. 2026

- Worked for **Dr. Tishara Garg** and **Dr. Edward Wiles** on the effects of India's GST reform on interstate trade
- Recovered and cleaned corrupted Karnataka VAT records and matched firm identifiers across the legacy VAT and GST systems, making it possible to follow the same firms through the reform
- Constructed time-consistent district and pincode crosswalks for India, reconciling administrative boundary splits and mergers so firm-level tax records could be matched to a stable geography and to bilateral distance measures across the reform period
- Replicated buyer-seller relationship dynamics in Karnataka e-way bill data (value growth and survival curves estimated by PPML) to establish that the model's micro predictions hold in this setting before extending it
- Wrote a transition-path solver in Julia for a dynamic Eaton-Kortum model of trade cost shocks

Research Assistant | *Allard School of Law*

Sep. 2025 – Present

- Working for **Dr. Wei Cui** on empirical legal research on the history of international tax law
- Built an LLM coding pipeline extracting structured doctrinal variables from historical treaty texts and validated it against manual coding, making full-population coding feasible
- Converted scanned archival documents into machine-readable text, producing cleaner and more consistent source material than manual transcription of the same volume would have allowed

Research Assistant | *Vancouver School of Economics*

May 2025 – Aug. 2025

- Worked for **Dr. Michael Peters** on projects studying doctoral placement outcomes and hiring networks
- Represented doctoral placement flows as a directed hiring network and applied graph-theoretic measures of centrality, documenting within-institution hiring consistent with network effects rather than rank-based sorting
- Built a recruitment dashboard used by **EconJobMarket** to track 16,500+ applicants across PhD cohorts

Mathematics Teaching Assistant | *UBC Department of Mathematics*

Aug. 2024 – Present

- Lead weekly lectures and problem-solving sessions for a 50+ student introductory calculus course
- Ranked among the top TAs in the department with a 99%+ student satisfaction rating ([TA Report](#))

Software Developer Intern | *iCliniq*

May 2023 – Jul. 2023

- Fine-tuned open medical language models (Llama-2, BioMedLM, RedPajama) on proprietary clinical records and Q&A data to adapt general-purpose models to the company's patient-facing domain
- Applied QLoRA and PEFT so adaptation could run on available GPU hardware without full-model retraining

SKILLS & INTERESTS

Technical Skills: Claude Code, Python, R, Stata, Julia, Java, C/C++, SQL, GitHub and L^AT_EX

Hobbies & Interests: Boulderling, Biking, Hiking, Football, Pickleball and Film-making